R210517

Q.P. Code: 2412075/A

R-23 **Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh**
E-1 Sem-2 End Semester Examinations, June-24
23MA1201-Linear Algebra & Numerical Methods
(Common to ECE, EEE, MME, CHE, ME, CE)

Time: 3 hours **I.D No.:** R210517 **Max Marks: 60**

Part – A

Answer all the following Questions. Each question carries one mark. 18 X 1 = 18 Marks

1 Which of the following set of vectors is linearly dependent over their respective vector space ?

A $\{u_1 = (1,2,5), u_2 = (2,5,1), u_3 = (1,5,2)\} \in \mathbf{R}^3(\mathbf{R})$

B $\{f(x) = \sin x,\ g(x) = \cos x,\ h(x) = x\} \in \mathbf{R}(\mathbf{R})$

C $\{u = (1+i,\ 2i), v = (1,\ 1+i)\} \in \mathbf{C}^2(\mathbf{R})$

D $\left\{w_1 = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix},\ w_2 = \begin{bmatrix} 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}\right\} \in M_{2,3}(\mathbf{R})$

2 Consider the following subsets W_1 and W_2 of $V(\mathbf{R}^3)$ and $V(\mathbf{R}^4)$ respectively:

$$P : W_1 = \{a, b \in \mathbf{R} \mid (a, a^2, b)\}$$
$$Q : W_2 = \{(a,b,c,d) \in \mathbf{R}^4 \mid a - b - 3d = 0\}$$

Then

A W_1 is a subspace of $V(\mathbf{R}^4)$ and W_2 is not a subspace of $V(\mathbf{R}^3)$

B W_1 is not a subspace of $V(\mathbf{R}^4)$ and W_2 is a subspace of $V(\mathbf{R}^3)$

C W_1 is a subspace of $V(\mathbf{R}^4)$ and W_2 is a subspace of $V(\mathbf{R}^3)$

D W_1 is not a subspace of $V(\mathbf{R}^4)$ and W_2 is not a subspace of $V(\mathbf{R}^3)$

3 The coordinate vector of the matrix $A = \begin{bmatrix} 2 & 3 \\ 4 & -7 \end{bmatrix}$ in the real vector space $\mathbf{M}_{2,2}(\mathbf{R})$ relative to the basis $B = \left\{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right\}$ is

A $[A]_B = [-7, 11, -21, 30]^T$

B $[A]_B = [7, -11, 21, -30]^T$

C $[A]_B = [-7, -11, 21, -30]^T$

D $[A]_B = [7, 11, -21, 30]^T$

4 Consider the matrix $A = \begin{bmatrix} 0 & 1 & -1 \\ -6 & -11 & 6 \\ -6 & -11 & 5 \end{bmatrix}$. Then the absolute value of the ratio of the maximum Eigen value to the minimum Eigen value is

A 3

B 2

C $\frac{1}{3}$

D $\frac{1}{2}$

5 For what value of a , the system of linear equations $x + ay + z = 3,\ x + 2y + 2z = 5,\ x + 5y + 3z = 9$ has no solution?

A -1

B 0

C 1

D 2

6 If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then the sum of the Eigen values of the matrix A^5 is

A 276

B 243

C 216

D 186

7 The volume of the region bounded above by the elliptical paraboloid $\mathbf{z = 10 + x^2 + 3y^2}$ and below by the rectangle $\mathbf{R : 0 \le x \le 1, 0 \le y \le 2}$ is

A $\frac{28}{3}$

B $\frac{86}{3}$

C $\frac{68}{3}$

D $\frac{82}{3}$

8 $\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2}(x^2+y^2)\,dzdydx =$

A $\frac{4\pi}{25}$ B $\frac{8\pi}{5}$

C $\frac{12\pi}{25}$ D $\frac{16\pi}{5}$

9 The area of the region enclosed by the astroid $\mathbf{r}(t) = (\cos^3 t)\,\mathbf{i} + (\sin^3 t)\,\mathbf{j},\ 0 \le t \le 2\pi$ is

A $\frac{2\pi}{3}$ B $\frac{\pi}{4}$

C $\frac{3\pi}{8}$ D $\frac{5\pi}{8}$

10 Using Green's theorem area formula, the area of the following astroid is

$$\mathbf{r}(t) = \cos^3 t\ \mathbf{i} + \sin^3 t\ \mathbf{j}, 0 \le t \le 2\pi$$

A $\frac{3\pi}{4}$ B $\frac{3\pi}{8}$

C $\frac{4\pi}{3}$ D $\frac{8\pi}{3}$

11 The normal vector to the surface $\mathbf{xy^2 + 2yz = 8}$ at the point $(3, -2, 1)$ is

A $\mathbf{i} - 10\,\mathbf{j} + \mathbf{k}$ B $4\,\mathbf{i} + \mathbf{j} - 4\,\mathbf{k}$

C $10\mathbf{i} + 10\,\mathbf{j} + 4\,\mathbf{k}$ D $4\,\mathbf{i} - 10\,\mathbf{j} - 4\,\mathbf{k}$

12 Using Gauss-Divergence theorem, the flux of $\mathbf{F} = \mathbf{xy\,i + yz\,j + zx\,k}$ outward through the surface of the cube cut from the first octant by the planes $x = 1, y = 1$ and $z = 1$ is

A $\frac{1}{2}$ B $\frac{2}{3}$

C $\frac{3}{2}$ D $\frac{3}{4}$

13 Let Δ and ∇ denotes forward and backward difference operators respectively and consider the following statements:

$$S_1 : \nabla - \Delta = -\Delta\nabla \quad ; \quad S_2 : \Delta - \nabla = \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$$

Then

A is true and S_2 is false B S_1 is false and S_2 is true

C S_1 is true and S_2 is true D S_1 is false and S_2 is false

14 The rate of convergence of the Regula-Falsi Method is

A 1 B 1.62

C 1.78 D 2

15 The number of iterations required to find a root of the equation $\mathbf{x^3 - 5x + 1 = 0}$ which lies i between 0 and 1 with 0.004 accuracy by using Bisection method is

A 6 B 8

C 10 D 12

16 Using the following data and the Trapezoidal Rule with $n = 3$, the approximate value of the integral $\int_1^4 f(x)\ dx$ is

x	1	2	3	4
f(x)	0.3679	0.1353	0.0498	0.0183

A 0.3986 B 0.3782

C 0.3525 D 0.3379

17 The error bound in evaluating the integral $\int_0^1 \frac{1}{3+2x}\ dx$ using Simpson's $\frac{1}{3}$ Rule with $n = 4$ is

A 0.00343 B 0.000443

C 0.0000443 D 0.0000343

18 Using Simpson's $\frac{3}{8}$ Rule with 4 nodal points, the approximate value of the integral $\int_1^2 \frac{1}{5+3x}\ dx$ is

A 0.10696 B 0.10686

C 0.10636 D 0.10616

6 Evaluate the integral $\iiint_D xe^{x^2+y^2+z^2}\, dV$, where D is the portion of the unit ball $x^2+y^2+z^2 \leq 1$ that lies in the first octant. 7M

UNIT-4

7 Show that the line integral $\int_C 2\cos y\, dx + \left(\frac{1}{y} - 2x\sin y\right) dy + \frac{1}{z} dz$ is independent of the path of integration from $(0,2,1)$ to $\left(1, \frac{\pi}{2}, 2\right)$. Evaluate the integral. 7M

(OR)

8 Verify the Stokes' theorem for the vector field $F = (3x - y)\, \boldsymbol{i} - 2yz^2\, \boldsymbol{j} - 2y^2 z\, \boldsymbol{k}$, where S is the surface of the sphere $x^2+y^2+z^2 = 16, z > 0$. 7M

UNIT-5

9 Use Newton-Raphson Method to obtain a root correct up to 3 decimal places of the equation $e^x \sin x - x^2 = 0$, with an initial approximation $x_0 = 3$. 7M

(OR)

10 Consider the following tabulated points to compute the value of $f(1.1)$ using Newton's difference interpolating formula. 7M

x	1.0	1.2	1.4	1.6	1.8	2.0
$f(x)$	2.71828	3.32012	4.05520	4.95303	6.04965	7.38906

UNIT-6

11 Evaluate the integral $\int_0^4 \sqrt{3+\sqrt{x}}\, dx$ using Simpson's Rule with $n = 8$ correct up to 6 decimal places. 7M

(OR)

12 Using Euler's Method to obtain the approximate value of $y(0.5)$ correct upto 5 decimal places for the IVP $y' = (x-y)^2,\ y(0) = 0.5$ with step size $h = 0.1$. 7M

***** THE END *****

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Q.P. Code: 2412075/A

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh
E-1 Sem-2 End Semester Examinations, June-24
23MA1201-Linear Algebra & Numerical Methods
(Common to ECE, EEE, MME, CHE, ME, CE)

Time: 3 hours I.D No.: R210517 Max Marks: 60

Part – B

Answer all the following Questions. Each question carries seven marks 6X7= 42 Marks

UNIT-1

1 Show that $B = \{(1,0,1),(2,1,2),(1,2,2)$ is basis of R^3. If E is the standard basis of R^3, then find the transition matrix P from the standard basis E to B and the transition matrix Q from B to E. Then verify that $Q = P^{-1}$ or not? 7M

(OR)

2 If $T : R^4 \longrightarrow R^3$ is a linear transformation defined by 7M

$T(a,b,c,d) = (a + 2b + 3c + 2d,\ 2a + 4b + 7c + 5d,\ a + 2b + 6c + 5d)$,

where $a,b,c,d \in R$, then find the basis and dimension of the $Ker(T)$ and $Range(T)$, and hence verify Rank-Nullity Theorem.

UNIT-2

3 Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 0 & i & i \\ i & 0 & i \\ i & i & 0 \end{bmatrix}$. 7M

Instruction: Zero marks will be awarded if the Eigen values or Eigen vectors are computed using the calculator or any electronic gadget without showing proper calculated steps.

(OR)

4 Find the inverse of the matrix A using Cayley-Hamilton Theorem. 7M

$$A = \begin{bmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

UNIT-3

5 Find the moments of inertia of mass about the coordinate axes and origin of a triangular lamina with vertices $(0,0),(1,0)$ and $(0,2)$ if the density function is $\rho(x,y) = 1 + 3x + y$. 7M

(OR)

Q.P. Code: 2412001/A

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh
E-1 Sem-2 End Semester Examinations, June-24

23BE1201 Environmental Science
(Common to CE, ME, CSE)

ABIG14

27

Time: 3 hours Student I.D No.: R210517. Max Marks: 100

Part – A

Answer all the following Questions. Each question carries one mark. 28 X 1 = 28 Marks

1 World Environment Day is celebrated on

A 10th June B 5th July

C 5th June D 15th June

2 The World Summit on Sustainable Development was held at

A Cape town B Rio de Janerio

C New Jersey D Johannesberg

3 GIS Stands for

A Geographic Information Systems B Geographic Indian Systems

C Geological Information Systems D Geological Industrial Systems

4 Natural Geysers which operate due to geothermal energy are present in

A Manikaran in Kullu B Sohana in Haryana

C Karnol in Haryana D

5 Environmental activist Medha Patkar has taken up issues related to which dam

A Bhakranangal B Sardar Sarovar

C Nagarjuna Sagar D Govinda Sagar

6 Biomass energy can be obtained from

A Energy Plantations B Petro crops

C Agricultural and urban waste biomass D All of these

7 Pyramid of ---------- is always upright

A Number B Biomass

C Energy D None of the above

8 The term ecosystem was coined by

A Hatch B Tansley

C Calvin D Brown

9 The organisms who feed directly on producers are called

A Herbivores B Secondary consumers

C detritivores

D cornivores

10 The darker zone in lakes where light penetration is negligible is called

A Littoral zone

B Limnetic zone

C Profundal zone

D Euphotic zone

11 The progressive accumulation of some non-biodegradable chemicals through the food chain is known as

A Ecological balance

B Biological magnification

C Trophic structure

D Bio-degradation

12 The type of succession occurring on a bare rock is called

A Halosere

B lithosere

C hydrosere

D Sammosere

13 Which one of the following National Parks do not have tigers as their main wildlife

A Gir

B Corbett

C Dudwa

D Ranathambore

14 Kaziranga National Park is famous for

A One horned rhino

B Hangul

C Tiger

D Elephant

15 Cryopreservation of plant seeds and pollen is done at a very low temperature of -196°C by using

A Ice

B Ammonia

C Liquid Nitrogen

D Methane

16 Illegal killing of prohibited endangered animals is called

A culling

B Poaching

C Cannibalism

D All of the above

17 Species restricted only to a particular area are called

A Hotspot

B Pandemic

C Fragmentation

D Endemic

18 Which of the following is an example of ex-situ conservation

A Biosphere reserve

B Sanctuary

C National park

D Gene bank

19 Bhopal gas tragedy occurred due to leakage of

A Methyl Isocyanide

B DDT

C SO_2

D Dioxins

20 Itai Itai disease in Japan was caused by consumption of rice contaminated with

A Cobalt

B Mercury

C Chromium

D Cadmium

21 The most important air pollutant is

A SO_2 B NO_2

C CO_2 D RADON GAS

22 Blue baby syndrome is caused by the presence of ---------- in drinking water

A Nitrates B Cobalt

C Iron D Fluorides

23 As per the forest act, cultivation of which of the following is a non-forest activity

A Tea B Mulberry

C Rubber D All of these

24 Which article in constitution recognizes environmental protection as one of the fundamental duties of every citizen of India

A Article 42 B Article 48 A

C Article 51 A(g) D Article 52

25 Ecotourism involves the following elements

A Conservation of Biodiversity B Promotion of local people involvement

C Ecological balance D All of the above

26 Ghana Bird sanctuary is located in

A Rajasthan B J&K

C Maharashtra D Haryana

27 Gasohol is a mixture of

A Ethanol and gasoline B Ethanol and methane

C Methane and gasoline D Gasoline and ammonia

28 Eutrophication of lakes is caused the excessive presence of

A Nitrates and chlorides B Nitrates and phosphates

C Chlorides and sulphates D Ca and Mg

Q.P. Code: 2412001/A

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh
E-1 Sem-2 End Semester Examinations, June-24

23BE1201 Environmental Science
(Common to CE, ME, CSE)

Time: 3 hours Student I.D No.: Max Marks: 100

Part – B

Answer all the following Questions. Each question carries twelve marks 6X12= 72 Marks

UNIT-1

1 (a) Explain the scope and importance of Environmental studies? 8M
(b) Explain about the water cycle? 4M

(OR)

2 (a) Explain the advantages and disadvantages of Solar Energy? 6M
(b) Discuss about Deforestation? 6M

UNIT-2

3 What are ecological pyramids? Explain some of these pyramids are upright while others are inverted in different ecosystems? 12M

(OR)

4 (a) What are the abiotic and biotic components in ecosystem? 8M
(b) Define the food chain and explain the types of food chains? 4M

UNIT-3

5 (a) What are the major threats to biodiversity? 6M
(b) Explain any 3 values of biodiversity? 6M

(OR)

6 Explain the conservative strategies of biodiversity? 12M

UNIT-4

7 Define the air pollution and explain the sources, effects and control measures of air pollution 12M

(OR)

8 (a) Explain the sources and effects of solid wastes 6M
(b) Discuss about cyclones 6M

UNIT-5

9 (a) Define the acid rains and explain the effects & control measures of acid rains? 6M
(b) Explain the sustainable development 6M

(OR)

10 (a) Explain the water (Prevention and Control of Pollution) act 1974. 6M
(b) What is a climate change and explain the key factors for climate change? 6M

UNIT-6

11 Explain the human population and environment 12M

(OR)

12 (a) Explain water related diseases 6M
(b) Urban poverty and environment 6M

***** THE END *****

R210517

Q.P. Code: 2412142/A

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh
E-1 Sem-2 End Semester Examinations, June-24

23CS1265 Advanced Programming Course
(Civil Engineering)

Time: 3 hours Student I.D No.: Max Marks: 60

Part – A

Answer all the following Questions. Each question carries one mark. 18 X 1 = 18 Marks

1 How is an array initialized in C language?

A int a[3] = {1, 2, 3};
B int a = {1, 2, 3};
C int a[] = new int[3]
D int a(3) = [1, 2, 3];

2 Which of the following is a linear data structure?

A Array
B AVL Tress
C Binary Tress
D Graphs

3 What is the output of the following code snippet?

```
void solve() {
 stack<int> s;
 s.push(1);
 s.push(2);
 s.push(3);
 for(int i = 1; i <= 3; i++) {
   cout << s.top() << " ";
   s.pop();
 }
}
```

A 321
B 123
C 3
D 1

4 Which one of the following is an application of queue data structure

A When a resource is shared among multiple consumers.
B (a) When data is transferred asynchronously
C (a) Load Balancing
D (a) All of the above

5 When a pop() operation is called on an empty queue, what is the condition called?

A Overflow
B Underflow
C Syntax Error
D Garbage Value

6 Which of the following data structures finds its use in recursion?

A Stack
B Arrays
C Linked List
D Queues

7 What is a data structure?

A A programming language
B A collection of algorithms
C A way to store and organize data
D A type of computer hardware

8 **What is the value of the postfix expression 6 3 2 4 + - *?**

A 74
B -18
C 22
D 40

9 **The data structure required for Breadth First Traversal on a graph is?**

A Array
B Stack
C Tree
D Queue

10 **Which data structure is based on the Last in First out (LIFO) principle?**

A Tree
B Linked List
C Stack
D Queue

11 **Which of the following application makes use of a circular linked list**

A Recursive function calls
B Undo operation in a text editor
C Implement Hash Tables
D Allocating CPU to resource

12 **Which of the following tree data structures is not a balanced binary tree?**

A Splay tree
B B-tree
C AVL tree
D Red-black tree

13 **Which of the following is the most widely used external memory data structure?**

A B-tree
B Red-black tree
C AVL tree
D Both AVL tree and Red-black tree

14 **What is an AVL tree?**

A a tree which is unbalanced and is a height balanced tree
B a tree which is balanced and is a height balanced tree
C a tree with at most 3 children
D a tree with three children

15 **What is the advantage of a hash table as a data structure?**

A Easy to implement
B faster access of data
C Exhibit good locality of reference
D very efficient for less number of entire

16 What is the time complexity of the binary search algorithm?

A O(n)
B O(1)
C O(log2n)
D O(n^2)

17 **What is the time complexity to insert an element to the front of a Linked List (head pointer given)?**

A O(n)
B O(1)
C O(logn)
D O(n * logn)

18 **Which of the following represents the Postorder Traversal of a Binary Tree?**

A Left -> Right -> Root
B Left -> Root -> Right
C Right -> Left -> Root
D Right -> Root -> Left

P2105

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh Q.P. Code: 2412142/A

E-1 Sem-2 End Semester Examinations, June-24

23CS1265 Advanced Programming Course
(Civil Engineering)

Time: 3 hours Student I.D No.: Max Marks: 60

Part – B

Answer all the following Questions. Each question carries seven marks 6X7= 42 Marks

UNIT-1

• 1 What is single linked list? 1M

Write an algorithm for the following single linked list operations

a. Display 3M

b. Insertion a node at beginning 3M

(OR)

2 Create a doubly linked list and write a program in C to insert and deletion a node at any position in a doubly linked list 1M 3M 3M

UNIT-2

3 Write a program in c implementation of stack using array for the following operations 2M

a. push() 2M

b. pop() 3M

c. display()

(OR)

• 4 Write a tabular method for infix to postfix for the fallowing expression 7M

K+L-M*N+ (O^P)*W/U/V*T+Q

UNIT-3

• 5 Write algorithm for the radix sort given array 7M

432	8	530	90	88	231	11	45	677	199

(OR)

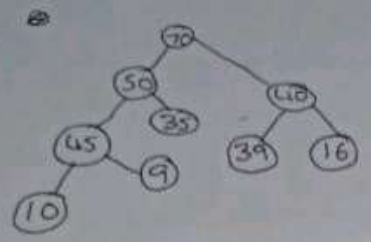
6 What is Hashing? Write the types of Hashing? Write hash table A=3,2,9,6,11,13,7,12 h(k)=2k+3 ,m=10,use division method & open addressing to store the values 2M 2M 3M

UNIT-4

• 7 Write tree terminology, Binary Tree Represent by using array and linked representations? 2M 2M 3M

(OR)

8 What is heap, the given max heap tree inserts an element 60, write code and draw the max heap tree after inserting the element



1M
3M
3M

UNIT-5

9 What is AVL tree? and Construct AVL tree for the fallowing elements 14,17,11,53,4,13,12,8,60,19,16,20 — 1M, 6M

(OR)

10 What is Binary Search tree ,draw search tree by inserting the fallowing numbers from left to right 11,6,8,19,4,10,5,17,43,49,31 — 1M, 6M

UNIT-6

11 What is graph, Write adjacency matrix & adjacency list below graph?

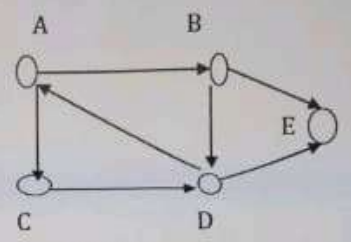
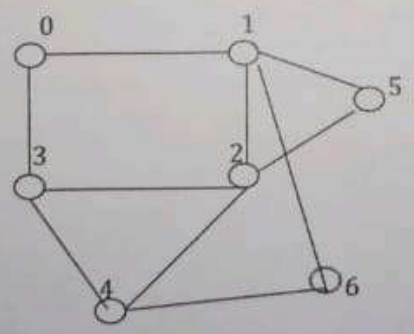


Figure — 7M

(OR)

12 Write the BFS and DFS of the above the graph — 7M



***** THE END *****

ABIHY

Q.P.Code: 2412115/B

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh
E-1 Sem-2 End Semester Examinations, June-24

23CE1202 Engineering Geology
(Civil Engineering)

Time: 3 hours Student-I.D No.: R210519 Max Marks: 60

Part - A

Answer all the following Questions. Each question carries one mark. 18 X 1 = 18 Marks

1 What role does weathering play in the context of dams and reservoirs?

A It strengthens the structure of dams
B It reduces the stability of reservoir banks
C It has no significant impact on water storage
D It increases the lifespan of tunnels

2 Which type of weathering is most commonly associated with the deterioration of granite?

A Mechanical weathering
B Chemical weathering
C Biological weathering
D Physical weathering

3 How does an understanding of weathering contribute to civil engineering projects?

A It helps in selecting appropriate construction materials
B It reduces construction costs
C It speeds up project completion
D It minimizes environmental impacts.

4 Which mineral is commonly used in making ceramics and glass?

A Feldspar
B Quartz
C Olivine
D Augite

5 Which of the following is a physical property of minerals?

A Color
B Taste
C Smell
D Sound

6 Which of the following minerals is commonly used in insulation due to its heat-resistant properties?

A Muscovite
B Biotite
C Asbestos
D Garnet

7 Conglomerate is primarily composed of:

A Fine-grained minerals
B Sand-sized particles
C Rounded pebbles and gravel
D Clay minerals

8 Which of the following rocks is primarily composed of calcium carbonate?

A Sandstone
B Gneiss
C Marble
D Slate

9 What type of rock forms through the metamorphism of limestone?

each technique would be most appropriate, considering environmental and economic factors.

(OR)

*8 Describe the concept of strike and dip in structural geology. Provide an example of how strike and dip measurements are used to interpret geological structures. 7M

UNIT-5

*9 Describe the principles and applications of seismic methods in geophysical studies. Choose a hypothetical geological scenario (e.g., locating a fault line, mapping subsurface structures) and outline a seismic survey plan, including the equipment and procedures involved. Justify your choices based on the geological context and survey objectives. 7M

(OR)

10 Design an integrated geophysical survey plan to investigate the potential for groundwater resources and seismic hazards in a coastal region. Explain the rationale behind your choice of geophysical methods and survey parameters. 7M

UNIT-6

*11 (a) Describe the significance of geophysical investigations in civil engineering projects. Provide two examples of how geophysical surveys can aid in site characterization. 4M

(b) Explain how geological considerations differ for dam reservoirs, tunnels, and road cuts. 3M

(OR)

12 (a) Define geological investigations and list three common techniques used in such investigations. 3M

(b) Explain the basic principle behind remote sensing techniques in geology. 4M

***** THE END *****

R210517.

Q.P. Code: 2412115/B

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh
E-1 Sem-2 End Semester Examinations, June-24

23CE1202 Engineering Geology
(Civil Engineering)

Time: 3 hours Student-I.D No.: R210517. Max Marks: 60

Part – B

Answer all the following Questions. Each question carries seven marks 6X7= 42 Marks

UNIT-1

1 (a) Discuss the significance of physical geology, petrology, and structural geology in civil engineering. 6M

(b) Provide examples of how each of these sub-disciplines contributes to the understanding and execution of civil engineering projects. 1M

2 (a) Analyze the weathering process of granite. Discuss the specific mechanisms involved in the weathering of granite. 4M

(b) How these processes can impact its suitability for use in civil engineering projects. 3M

UNIT-2

3 (a) Discuss any two methods used to study minerals and rocks. 4M

(b) Highlight their respective advantages and limitations. 3M

(OR)

4 (c) Discuss the economic significance of the any four common economic minerals. 4M

(d) Emphasize their industrial applications and extraction methods. 3M

UNIT-3

5 (a) Define the three main classifications of rocks in geology based on their formation. 1M

(b) Provide a brief explanation of each type. 6M

(OR)

6 Compare and contrast two sedimentary rocks of your choice in terms of their composition, formation processes, and distinguishing features. 7M

UNIT-4

7 Critically analyze the effectiveness of different soil stabilization techniques such as terracing, vegetation cover, and bioengineering. Provide examples of situations where 7M

A Gneiss
B Marble
C Schist
D Quartzite

10 What is the study of the orientation of geological structures such as folds, faults, and joints called?
A Petrology
B Structural Geology
C Geomorphology
D Sedimentology

11 Which of the following geological structures is characterized by a bending or curving of rock layers?
A Fault
B Joint
C Fold
D Conformity

12 The strike of a geological structure refers to:
A The direction of the slope
B The horizontal line on the plane of the structure
C The inclination of the structure
D The direction of the line formed by the intersection of the structure with a horizontal plane

13 Which of the following is NOT a method used in geophysical studies?
A Gravity methods
B Seismic methods
C Volcanic methods
D Electrical methods

14 Why is the study of groundwater important?
A It helps in predicting earthquakes
B It provides insights into climate change
C It is a vital source of drinking water and irrigation
D It aids in understanding plate tectonics

15 Which geophysical method is specifically useful for detecting changes in subsurface electrical resistivity?
A Gravity methods
B Magnetic methods
C Electrical methods
D Seismic methods

16 Which geophysical method is commonly used to detect subsurface features such as faults and bedrock variations?
A Ground-penetrating radar
B Seismic reflection
C Electrical resistivity imaging
D Magnetometry

17 Remote sensing techniques are utilized in geological investigations primarily for:
A Identifying mineral compositions
B Mapping surface topography
C Measuring ground temperature
D Analyzing groundwater quality

18 What geological considerations are crucial for the construction of dam reservoirs?
A Bedrock depth and stability
B Mineral composition of soil
C Groundwater flow rates
D Vegetation cover

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh

Q.P. Code: 2412109/ B

E-1 Sem-2 End Semester Examinations, June-24

23CE1201 Engineering Mechanics
(Civil Engineering)

Time: 3 hours Student-I.D No.: Max Marks: 60

Part – A

Answer all the following Questions. Each question carries one mark. 18 X 1 = 18 Marks

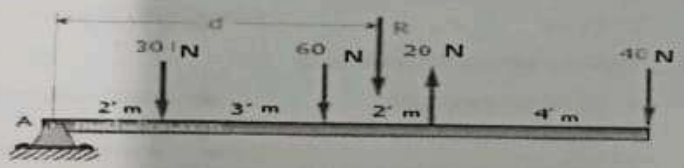
1 Two forces of magnitudes 4N and 5N at an angle of 60° the resultant force is equal to

A 4.5N
B 7.8N
C 6.4N
D 3N

2 When can two forces be in equilibrium?

A They are equal in magnitude
B They are collinear
C They are opposite in direction
D All the mentioned

3 A parallel force system acts on the liver arm. The magnitude and position of the resultant from A is

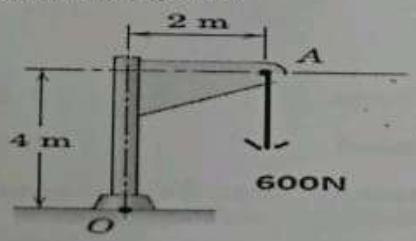


A 150N-5.15m
B 150N -5.5m
C 110N -5.5m
D 110N-6m

4 If several forces are acting at a point, their resultant will be inclined at an angle θ with the horizontal, such that.

A Sin θ = ΣV/ΣH
B Sin θ = ΣH/ΣV
C Tan θ = ΣH/ΣV
D Tan θ = ΣV/ΣH

5 The magnitude of the moment about the base point O is



UNIT-5

9 The motion of a particle is given as $a = 2t^3 - t^2 + 4$, where 'a' is acceleration and 't' is travelling time of particle. When t=1 sec, the displacement is 10 m and velocity is 8 m/sec. Find the displacement and velocity when t = 3 sec. 7M

(OR)

10 An aircraft is flying at an altitude of 2.5km with a uniform velocity of 650kmph. The pilot wants to release a bomb to hit the target on the ground at what horizontal distance from the target he should release the bomb to hit the target? 7M

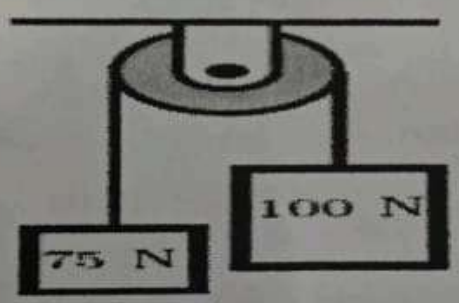
UNIT-6

11 A car is travelling at a speed of 80kmph suddenly reduced its speed and stops after skidding 40m. Determine 3M 4M

1) Time required to stop the car.
2) The coefficient of friction between the tire and the road surface

(OR)

12 Two blocks of weights 75N and 100N are hung as shown in the fig below. Assume pulley is friction less. How much distance the blocks will move while increasing their velocity from zero to 4m/s? What is the tension in the string? 7M



***** THE END *****

Q.P.Code: 2412109 B

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh
E-1 Sem-2 End Semester Examinations, June-24

23CE1201 Engineering Mechanics
(Civil Engineering)

Time: 3 hours Student-I.D No.: Max Marks: 60

Part – B

Answer all the following Questions. Each question carries seven marks 6X7= 42 Marks

UNIT-1

1 Find the resultant magnitude and direction of the force system shown in Fig. 7M

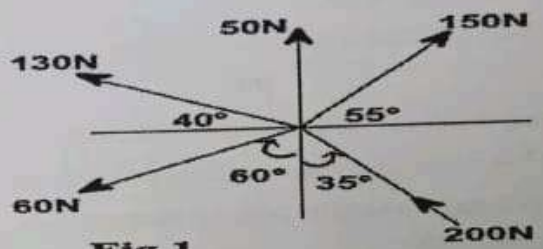
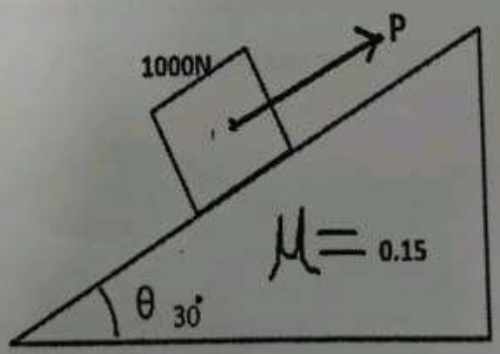


Fig.1

(OR)

2 (a) Define angle of friction and coefficient of friction 2M

(b) Block of weight 1000N is kept on an inclined plane (30 Degrees). Force P is applied parallel to the plane to keep the body in equilibrium. Determine the range of values of P for which the block will be in equilibrium. 5M



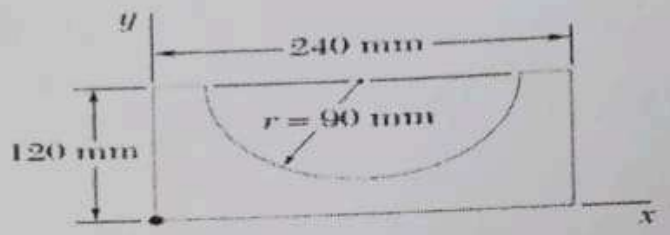
UNIT-2

3 Find the Centroid of the Channel-section shown in Fig. 7M

(OR)

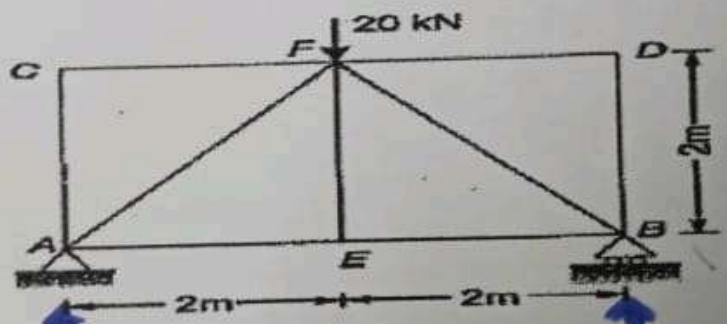
6 Find Moment of Inertia of the compound diagram shown in fig about its centroidal axis of the compound diagram (only I_{xx}). 7M

(All dimensions shown are in mm)



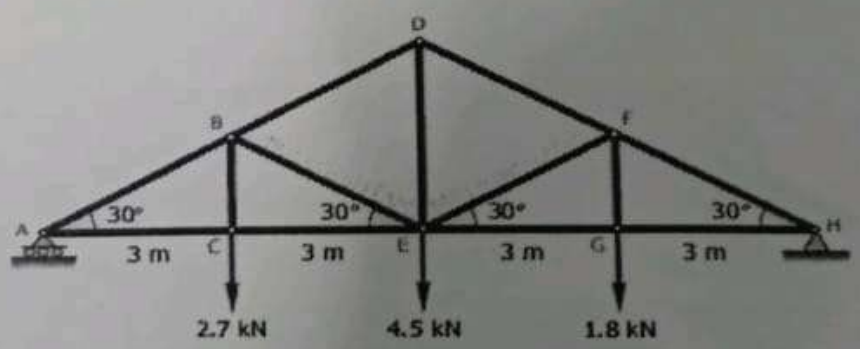
UNIT-4

7 Find the member forces in the trusses shown in fig by method of joints. 7M



(OR)

8 Using the method of sections determine the force in BD, BE, CE Members in the given truss. 7M



A 2400N

B 1200N

C 1800N

D ZERO

6 Which of the following is a feature of friction?

A Always acts in the direction opposite to the applied force

B Not a self-adjusting force

C Always acts in the direction of applied force

D It is an active force

7 Polar moment of inertia of a circular section diameter D is

A $\Pi D^4/64$

B $\Pi D^4/16$

C $\Pi D^4/32$

D $\Pi D^4/8$

8 The Centre of Gravity of a semi-circle lies at a distance _____ from its base.

A $4r/3\pi$

B $3r/4\pi$

C $2r/3\pi$

D $3r/2\pi$

9 Radius of gyration k relation with area(A) and moment of inertia(I) is

A $K^2 = I/A$

B $K^2 = I^2/A$

C $k = I/A$

D $K^2 = (I/A)^{1/2}$

10 Moment of Inertia is also known as the

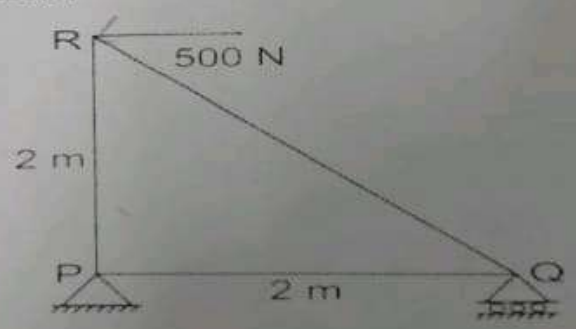
A Second moment of mass

B Second moment of Area

C Second moment of force

D All the above

11 For the given truss PQR, the force in the member PR



A 500(tensile)

B 500(compression)

C 650(tensile)

D 650(compression)

12 A bus travels with a speed of 20 m/s when accelerated at 10 m/s^2 from its rest position. What is the distance traveled?

A 10 m

B 15 m

C 20 m D 25 m

13 To take the longest possible jump, an athlete should make ana angle of

A 90^0 with the ground B 60^0 with the ground

C 45^0 with the ground D 30^0 with the ground

14 The rate of change of Velocity of a body is called.

A Acceleration B Distance

C Momentum D Displacement

15 What is the velocity of particle after 3 sec where its positioned is defined by $x=3t^2$ (x in metres)

A $18m/sec^2$ B $6m/sec^2$

C $3m/sec^2$ D $12m/sec^2$

16 Horizontal Distance travelled by a projectile is called

A Range B Distance

C Displacement D None of the above

17 the range of a projectile fired at ana angle of 15 is 50m. if it is fired with the same speed at ana angle of 45^0, its range will be

A 25m B 37m

C 50m D 100m

18 The unit of energy in S.I. units is

A Dyne B Kg

C Watt D Joule

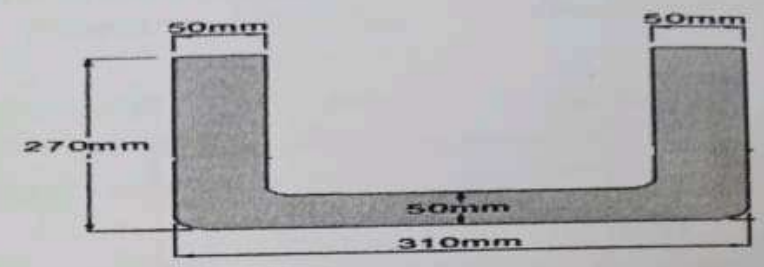
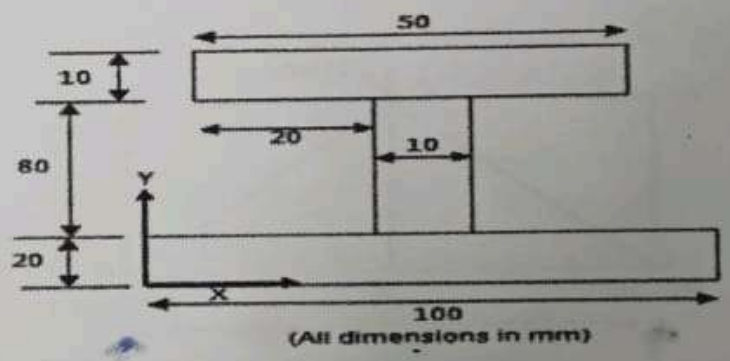


Fig.2

(OR)

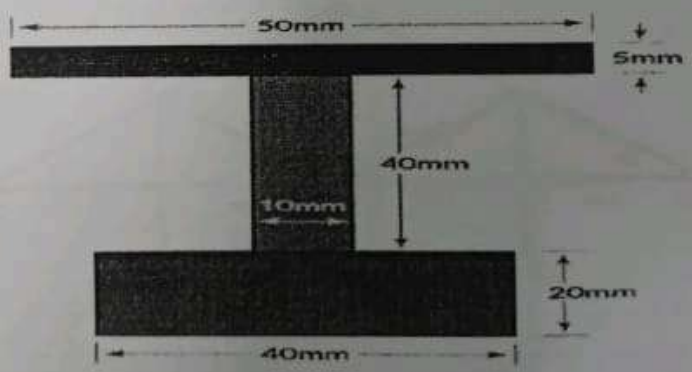
4 Locate the centroid of the I section shown in fig. 7M

(All dimensions shown are in mm)



UNIT-3

5 Find moment of inertia of I section shown in fig about its Centroidal axis (only I_{xx}) 7M



Q.P. Code: 2412068/A

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh
E-1 Sem-2 End Semester Examinations, June-24

23EE 1209 Basic Electrical and Electronics Engineering
(Common to CE, MME)

Time: 3 hours Student I.D No.: R210517. Max Marks: 60

Part – A

Answer all the following Questions. Each question carries one mark. 18 X 1 = 18 Marks

1 A practical current source is usually represented by

A A resistance in series with an ideal current source
B A resistance in parallel with an ideal current source
C A resistance in series with an ideal voltage source
D A resistance in parallel with an ideal voltage source

2 Six incandescent bulbs are connected across 200v. Each bulb is rated at 100w. How much current flows through each bulb?

A 0.5A
B 2A
C 6A
D 7A

3 Superposition theorem is not valid for

A Current responses
B Voltage responses
C Linear response
D Power responses

4 What is the instantaneous value of $V(t)=5\ Sin(2p \times 10^4 t)$ volts at time, $t=25\mu$ sec

A -10V
B 5V
C 10V
D 0.137V

5 The ratio of RMS value to the average value is called _____

A Peak factor
B Form factor
C Power factor
D All

6 The form factor of a sinusoidal waveform is _____

A 1.11
B 1.414
C 1.732
D 0

7 The commutator in a DC machine can be convert

A AC to DC
B DC to AC
C Both A & B
D None

8 Why is the armature of a DC machine made of silicon steel stampings?

A To reduce eddy current loss
B To reduce hysteresis loss
C For the ease with which the slots can be created
D To achieve high permeability

9 In shunt motor armature current is equal to

A Load current

B The sum of motor current and field current

C The difference of motor current and field current

D Both motor and field current

10 A transformer works on

A DC

B AC

C AC & DC Both

D Neither AC or DC

11 The frequency of the secondary voltage of a transformer will be _____

A Less than the frequency of the primary voltage

B Equal to the primary voltage

C Greater than the frequency of the primary voltage

D Very much greater than the frequency of the primary voltage

12 Difference in speed between stator field and rotor is called

A Full load speed

B No load speed

C slip

D regulation

13 The diode _____

A Is the simple of semiconductor devices

B Has characteristics that closely match those of a simple switch

C Is a two terminal device

D All of the above

14 The output frequency of a full-wave rectifier is _____ the input frequency

A One-half

B Equal to

C Twice

D One-quarter

15 In a ZENER regulator, the change in load current produces , change in

A Zener voltage

B Zener current

C Both A & B

D None

16 A PNP transistor is made of

A Silicon

B Germanium

C Either silicon or germanium

D None

17 Which transistor configuration is known for its high input impedance and low output impedance, making it suitable for impedance matching?

A Common emitter

B Common base

C Common collector

D Darlington pair

18 The common base (CB) configuration is characterized by :

A High voltage gain and low current gain

B High current gain and low voltage gain

C Equal voltage and current gains

D No voltage or current gain

Q.P. Code: 2412068/A

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh
E-1 Sem-2 End Semester Examinations, June-24

23EE 1209 Basic Electrical and Electronics Engineering
(Common to CE, MME)

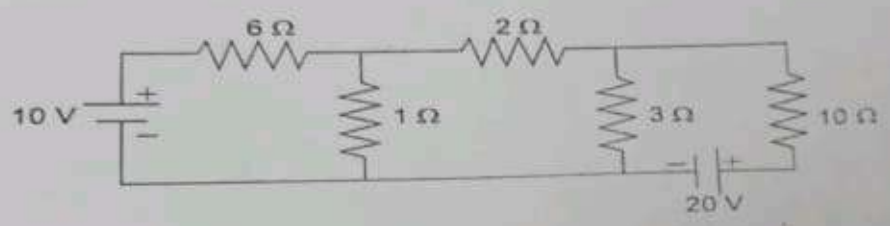
Time: 3 hours Student I.D No.: R210517 Max Marks: 60

Part - B

Answer all the following Questions. Each question carries seven marks 6X7= 42 Marks

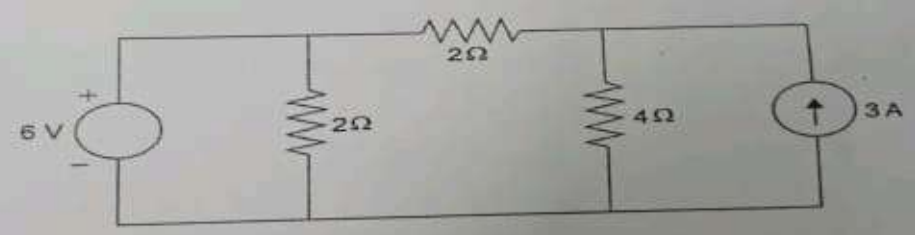
UNIT-1

1 Find the mesh currents for the circuit shown in the figure. 7M



(OR)

2 Determine the current through resistor 4Ω using superposition theorem as shown in figure 7M



UNIT-2

3 (a)The voltage and current across a load are given by V=110V and I=20∠-50° A. Find active power, reactive power and apparent power. 3M

(b)The equation of an alternating current is i(t)=42.42sin(628t). Determine (i)RMS value (ii) Average value (iii) Form factor (iv) Peak factor 4M

(OR)

4 In case of pure inductor, the current lags the voltage by 90°. Justify. 7M

UNIT-3

5	(a)Derive the EMF equation of a DC generator	4M
	(b)Classify DC generators based on excitation	3M
	(OR)	
6	Briefly explain the three point starter with a neat sketch	7M
	UNIT-4	
7	Define efficiency and derive the condition for maximum efficiency of a single phase transformer	7M
	(OR)	
8	(a)Write the differences between squirrel cage and slip ring induction motor	5M
	(b) Define slip and slip speed	2M
	UNIT-5	
9	Explain the operation of PN junction diode	7M
	(OR)	
10	Explain the operation of Bridge Rectifier	7M
	UNIT-6	
11	Explain the operation of PNP Transistor	7M
	(OR)	
12	Explain how Transistor as an Amplifier	7M

***** THE END *****

R-23

R210181

Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh SET-C

E-1 Sem-1 End Semester Examinations, Dec-23

23EE1109 - Basic Electrical and Electronics Engineering

(Common to CSE & ME)

Time: 3 hours

Max Marks: 60

1 Number of independent meshes for a circuit with 6 branches and 3 nodes

A 2 B 3

C 4 D 5

2 Power dissipated across the resistor

A IR B IV

C IR^2 D I/R

3 The Norton impedence is ---------- the Thevenin impedence.

A Greater than B Same as

C Less than D Double than

4 Voltage lags current by 90° in which circuit

A Resistor Circuit B Capacitor Circuit

C Inductor Circuit D R-L Circuit

5 Form factor for sinusoidally varying quantity is = ----------

A 0.637 B 1.414

C 1.11 D 0.707

6 The average power supplied to an inductor over one complete alternating current cycle is....

A 0 B IV^2

C VI^2 D IR^2

7 Choice of material for DC Generator brushes

A Copper B Iron

C Carbon D Aluminium

8 Armature current I_a, In a three point starter = ...

A $E R_a/R_{st}$ B $E/R_{st}(1+R_a)$

C $E/(R_a+R_{st})$ D ER_{st}/R_a

9 Power developed by armature in Shunt Generator

A V_tI_L B EI_a

C $V_tI_L+ EI_a$ D $V_tI_L- EI_a$

10 Voltage transformation ratio (K) for a step up transformer

A K<1 B K>1

C K=1 D K=0

11 EMF equation of a transformer E= ----------. Φ_m=maximum flux in the core, f=frequency, N= Numer of turns

A $1.11\ f\Phi_m$

B $4.44fN\Phi_m$

C $4.44f\Phi_m$

D $4f\Phi_m$

12 If the synchronous speed is 'n_{sync}' and rotor speed is 'n_m', Slip S= ----------

A $n_m - n_{sync}$

B $n_{sync} - n_m$

C $1 - n_{sync}/ n_m$

D $1 - n_m/ n_{sync}$

13 A potential barrier of 0.50 V exists across a P-N junction. If the depletion region is $5X10^{-7}$ m wide, the intensity of the electric field in this region is

A $1X10^6$ V/m

B $1X10^5$ V/m

C $2X10^5$ V/m

D $2X10^6$ V/m

14 In comparison to a half wave rectifier, the full wave rectifier gives lower

A Efficiency

B Average dc

C Average output voltage

D None of these

15 Zener diode is used as

A Half wave rectifier

B Full wave rectifier

C ac voltage stabilizer

D dc voltage stabilizer

16 For a common base PNP transistor, current gain is 0.96 and the emitter current is 7.2 mA. The base current is--------

A 0.4 mA

B 0.2 mA

C 0.29 mA

D 0.35 mA

17 The transistors provide good power amplification when they are used in-------- configuration

A Common Collector

B Common Emitter

C Common Base

D None of these

18 For a transistor, the parameter β= 99. The value of the parameter α is

A 0.9

B 0.99

C 1

D 9

R-23 Rajiv Gandhi University of Knowledge Technologies-Andhra Pradesh

221018 1

E-1 Sem-1 End Semester Examinations, Dec-23

23EE1109 - Basic Electrical and Electronics Engineering

(Common to CSE & ME)

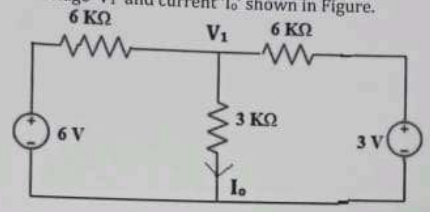
Time: 3 hours

Max Marks: 60

UNIT-1

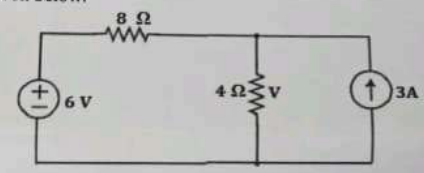
1 a). State and Explain Kirchoff's Current and Voltage laws. 7M

b). Find the nodal voltage 'V_1' and current 'I_o' shown in Figure.



(OR)

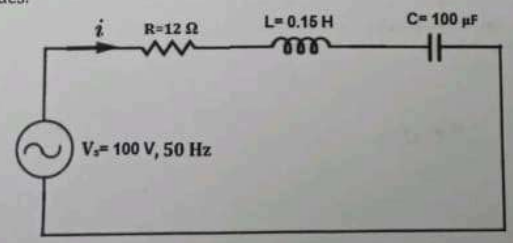
2 a). Calculate the voltage 'V' across the '4 Ω' resistor using Superposition theorem in the circuit given below. 7M



b). Show that the maximum power delivered from the source to the load resistor (R_L) occurs when R_{th}= R_L for DC circuits.

UNIT-2

3 A series RLC circuit containing a resistance of '12 Ω', an inductance of '0.15 H' and a capacitance of '100 μF' are connected in series with a 100 V, 50 Hz supply. Calculate total circuit impedence, the circuit current, voltage across each element and power factor values. 7M



(OR)

4 Compare Star & Delta connections in Three Phase system. 7M

UNIT-3

5 a). Derive the EMF equation of a DC generator. 7
b). Write the current and voltage relation for a shunt generator.

(OR)

6 a). Explain the Back EMF in DC motor. 7M
b). Derive the torque equation for a DC motor.

UNIT-4

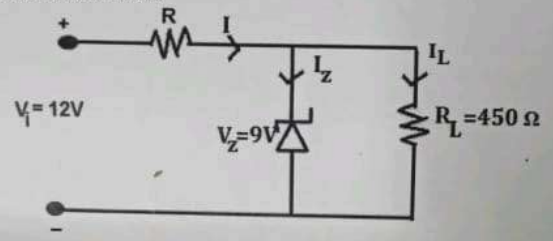
7 Discuss different kinds of losses in a transformer. Further explain the condition to achieve maximum efficiency of a transformer interms of losses. 7M

(OR)

8 Analyze the Torque-Slip characteristics of induction machine. 7M

UNIT-5

9 A 9V stabilized voltage supply is required to run a car stereo system from the cars 12V battery. A zener diode with V_z= 9V and P_{max}= 0.27 W is used as voltage regulator. Find the diode current and the value of the series resistor, if the load resistance is 450Ω. 7M



(OR)

10 Calculate the full wave rectifier efficiency by Mathematical Analysis. 7M

UNIT-6

11 Discuss the input and output characteristics of a transistor for common emitter configuration. 7M

(OR)

12 A transistor is connected in the CE configuration. The collector supply voltage is 10V and the voltage drop across the 500Ω resistor connected in the collector circuit is 0.6V. If α=0.96, find the a) collector emitter voltage, b) emitter current and c) base current. 7M

****** THE END *******